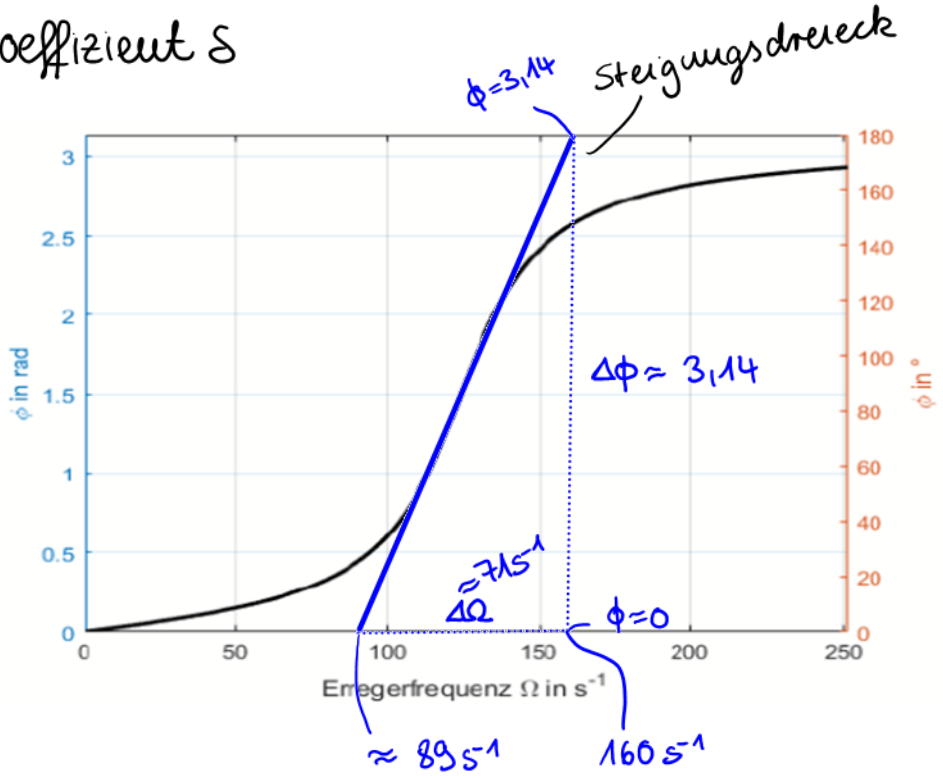
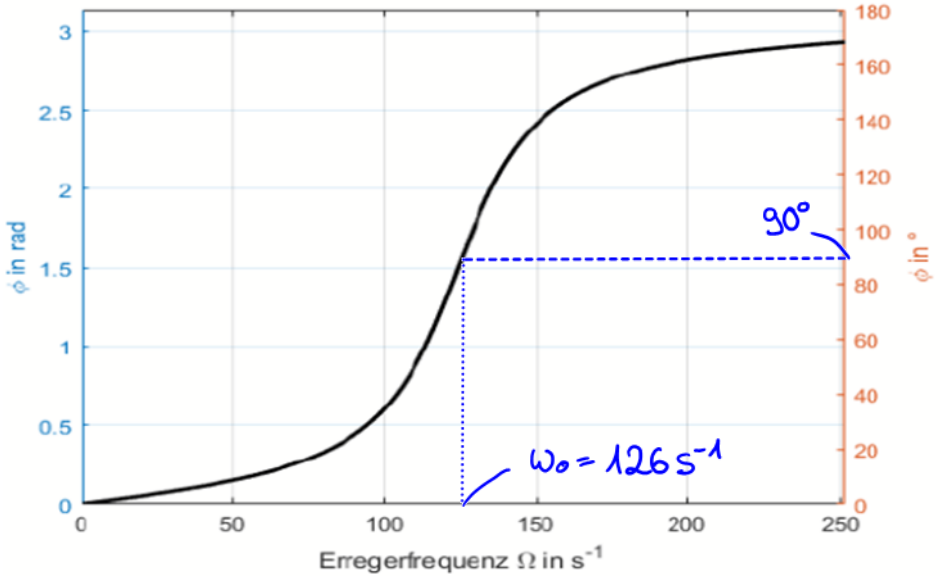
# Ausmessen des Phasengangs $\varphi(\omega)$

## a) Abklingkoeffizient $\delta$




$$\frac{1}{\delta} \approx \frac{\Delta\phi}{\Delta\Omega} \approx \frac{3{,}14}{71\,s^{-1}} \approx 0{,}044\,s \quad \Rightarrow \quad \underline{\underline{\delta \approx 22{,}6\,s^{-1}}}$$

- ungedämpfte Eigenfrequenz $\omega_0$




$$f_0 = \frac{\omega_0}{2\pi} \approx \frac{126\,s^{-1}}{6{,}24} \approx 20{,}2\,Hz$$

- Güte

$$Q \approx \frac{\omega_0}{2\delta} \approx \frac{126\,s^{-1}}{2 \cdot 22{,}6\,s^{-1}} \approx \underline{\underline{2{,}8}}$$